

Shubham Tanwar

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PROFESSIONAL SUMMARY

Software engineer with 2+ years of experience building and launching enterprise AI products at Boomi, supporting platforms serving 20,000+ customers and 25,000+ deployed AI agents. Experienced in React, TypeScript, Python, streaming AI interfaces, and cloud infrastructure. Currently building computer vision and 3D visualization systems at NYU.

SKILLS

Languages: Python, TypeScript, JavaScript, C++, SQL

AI/ML: PyTorch, NumPy, GenAI

Full-Stack: React, Redux, Jest/RTL, Node.js, FastAPI, Express, REST APIs, MySQL, NoSQL(MongoDB)

BigData & DevOps: HDFS, Apache Spark, AWS (Lambda, S3), Docker, CI/CD (Harness, CircleCI, Jenkins), Kubernetes

Competitive Programming: CodeChef 3★, 500+ LeetCode problems solved

WORK EXPERIENCE

Research Assistant

Sept 2026 - Present

AI4CE lab at NYU | Python, PyTorch, NumPy, Three.js

- Built an RGB-D reconstruction pipeline using **Python, NumPy, PyTorch, and SAM 3** to create 3D room scans and isolate the patient, bed, and C-arm.
- Developed an interactive **JavaScript and Three.js** application to align a virtual C-arm with scanned equipment and simulate its articulated motion.
- Implemented geometric collision detection against scanned patient and bed regions, creating interactive demonstrations that support ongoing research into camera-assisted C-arm positioning.

Software Engineer II (promoted after 14 months)

Mar 2023 - Jun 2025

Dell Boomi | TypeScript, React, Python, Redux, Jest/RTL, DevOps

- Built core discovery and installation experiences for Boomi Marketplace, **serving 20,000+ enterprise customers**, owned faceted search, solution detail pages, install flows, and frontend API contracts.
- Architected and shipped two **React** and **TypeScript** micro frontend applications for Boomi Agentstudio, taking Agent Garden and Agent Management from inception to early access launch in March 2025, the platform now supports 25,000+ deployed AI agents.
- Shipped a production integration dashboard with **React** and **Redux**, defined backend API contracts, and automated releases through **Harness** pipelines with **Snyk** and **SonarQube** security and quality checks.
- Engineered conversational AI interfaces with streaming responses, multi turn **state management**, tool execution visibility, and resilient **failure recovery**, including interrupted stream handling, retries, and optimistic UI rollback.
- Automated deployments with **CI/CD** pipelines (**Harness, AWS, YAML**) and built **Python** data pipelines using AWS Lambda, S3, and MySQL for telemetry tracking and event processing.
- Managed project milestones across concurrent engineering workstreams using **Agile** and Scrum practices, tracking delivery in **Jira**, maintaining organized technical records, and reporting progress, dependencies, and blockers to cross functional stakeholders.

PROJECTS

SVG Scaling Laws (ML / Transformers) | PyTorch, Python

- Trained GPT style transformers** from scratch across 1M to 86M parameters on 100M+ tokens, fitting scaling laws that predicted model performance with $R^2 = 0.96$.
- Built the training and evaluation pipeline including a custom **SentencePiece tokenizer**, distributed GPU training, and experiments comparing Standard and **Maximal Update Parameterization (μP)**.

Component-Based Jira Client | Python, FastAPI, AWS Lambda

- Built and deployed a serverless AI assistant that translates natural language requests into Jira operations through LLM tool calling, with **REST APIs**, **OAuth2** authentication, and **DynamoDB** session persistence.
- Designed extensible integrations using dependency injection and vendor neutral interfaces, enabling Slack connectivity and automated deployment through **CircleCI**.

EDUCATION

New York University

M.S., Computer Science, GPA: 3.7/4

Jan 2026 - Dec 2027

Guru Gobind Singh Indraprastha University

B.Tech, Information Technology, GPA: 8.3/10

June 2018 - Dec 2022

RELEVANT COURSES

- Deep Learning(Hons.), Machine Learning(Hons.), Big Data(Hons.), System Design**, Design & Analysis of Algorithms